University of Utah Spring 2026

MATH 2270-006
Final Questions

Instructor: Alp Uzman

April 27 2026, 8:00 AM - 10:00 AM

Surname:

First Name: KEY

uNID:

KEY

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1. **[40 points]** Consider the following system of linear equations:

$$x_1 + 3x_2 + 4x_3 + 5x_4 + 4x_5 = 0$$
$$2x_1 + 6x_2 + 9x_3 + 5x_4 + 2x_5 = 0$$

(a) **[10 points]** Write the system in matrix form.

$$A = \begin{pmatrix} 1 & 3 & 4 & 5 & 4 \\ 2 & 6 & 9 & 5 & 2 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \qquad \boxed{Ax = 0}$$

(b) **[10 points]** Solve the system.

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 4 \\ 2 & 6 & 9 & 5 & 2 \end{pmatrix} \to \begin{pmatrix} 1 & 3 & 4 & 5 & 4 \\ 0 & 0 & 1 & -5 & -6 \end{pmatrix} \to \begin{pmatrix} 1 & 3 & 5 & 0 & -2 \\ 0 & 0 & 1 & -5 & -6 \end{pmatrix}$$

$$\downarrow$$

$$\begin{pmatrix} 1 & 3 & 0 & 25 & 28 \\ 0 & 0 & 1 & -5 & -6 \end{pmatrix}$$

$$x_1 + 3x_2 + 25x_4 + 28x_5 = 0$$
$$x_3 - 5x_4 - 6x_5 = 0$$

$$\Downarrow$$

$$\boxed{\begin{aligned} x_1 &= -3x_2 - 25x_4 - 28x_5 \\ x_3 &= 5x_4 + 6x_5 \\ & x_2, x_4, x_5 \text{ free.} \end{aligned}}$$

(c) **[10 points]** Compute the dimension of the space of solutions.

# of frees = 3

(d) **[10 points]** Compute a five-dimensional vector orthogonal to any solution of the system.

nonzero.

$Sol = Nul(A)$

$Sol^{\perp} = Nul(A)^{\perp} = Col(A^T)$

$\Rightarrow \boxed{\begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \\ 4 \end{pmatrix} = 1\text{st row of } A \text{ works.}}$

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2. **[10 points]** Consider the following matrices:

$$A = \begin{pmatrix} 2 & 0 & -1 \\ 4 & -3 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{pmatrix}, \quad P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Compute the integers a, b, c, d, e, f such that

$$2PA - 3B = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

$$2PA - 3B = 2\begin{pmatrix} 4 & -3 & 2 \\ 2 & 0 & -1 \end{pmatrix} - 3\begin{pmatrix} 7 & -5 & 1 \\ 1 & -4 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} 8 & -6 & 4 \\ 4 & 0 & -2 \end{pmatrix} + \begin{pmatrix} -21 & 15 & -3 \\ -3 & 12 & 9 \end{pmatrix}$$

$$= \left(\begin{array}{c|c|c} -13 & 9 & 1 \\ \hline 1 & 12 & 7 \end{array}\right)$$

$\Rightarrow$

$$\boxed{\begin{aligned} a &= -13 \\ b &= 9 \\ c &= 1 \\ d &= 1 \\ e &= 12 \\ f &= 7 \end{aligned}}$$

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3. **[10 points]** Compute the rank of the following 4×5 matrix:

$$A = \begin{pmatrix} 2 \\ -2 \\ 3 \\ 5 \end{pmatrix} \begin{pmatrix} 23 \\ 17 \\ 31 \\ -2 \\ -7 \end{pmatrix}^T .$$

v

Outer product of nonzero vectors $\Rightarrow$ $\boxed{\text{rank} = 1}$

$$A = \begin{pmatrix} - & 2 & v & - \\ - & -2 & v & - \\ - & 3 & v & - \\ - & 5 & v & - \end{pmatrix} \rightarrow \begin{pmatrix} - & v & - \\ - & 0 & - \\ - & 0 & - \\ - & 0 & - \end{pmatrix}$$

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4. **[10 points]** Compute the determinant of the following matrix:

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 1 & 2 & -1 \\ 3 & -1 & 1 \end{pmatrix}.$$

$$\left(2\cdot 2\cdot 1 + 3\cdot 1\cdot(-1) + 3\cdot 1\cdot(-1)\right)$$

$$-\left(3\cdot 2\cdot 3 + 1\cdot 1\cdot 1 + 2\cdot(-1)\cdot(-1)\right)$$

$$= 4 - 3 - 3$$

$$- 18 - 1 - 2$$

$$= 2 - 7 - 18 = 2 - 25$$

$$= -23$$

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5. **[10 points]** Compute an orthonormal basis of the columnspace of the following matrix:

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -6 \end{pmatrix}.$$

$a_1 = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$ $a_2 = \begin{pmatrix} 2 \\ 3 \\ -6 \end{pmatrix}$.

$\langle a_1, a_1 \rangle = 9 + 4 + 4 = 17 \Rightarrow \boxed{|a_1| = \sqrt{17}}$

$\langle a_1, a_2 \rangle = 6 + 6 - 12 = 0 \quad \boxed{a_1 \perp a_2}$

$\langle a_2, a_2 \rangle = 4 + 9 + 36 = 49 \Rightarrow \boxed{|a_2| = 7}$

$\Rightarrow \left(u_1 = \frac{a_1}{|a_1|}, u_2 = \frac{a_2}{|a_2|} \right)$ ONB. of Col(A)

$u_1 = \begin{pmatrix} 3/\sqrt{17} \\ 2/\sqrt{17} \\ 2/\sqrt{17} \end{pmatrix}$ $u_2 = \begin{pmatrix} 2/7 \\ 3/7 \\ -6/7 \end{pmatrix}$.

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6. **[10 points]** Compute the positive singular values of the following matrix:

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & -3 & 2 \end{pmatrix}.$$

$$AA^T = \begin{pmatrix} 2 & 1 & -1 \\ 4 & -3 & 2 \end{pmatrix} \begin{pmatrix} 2 & 4 \\ 1 & -3 \\ -1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} 4+1+1 & 8-3-2 \\ & 16+9+4 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 29 \end{pmatrix}$$

$$(\lambda-6)(\lambda-29)-9$$

$$= \lambda^2 - 35\lambda + 29\cdot 6 - 9$$

$$= \lambda^2 - 35\lambda + 165$$

$$\begin{array}{r} 29 \\ \times\ 6 \\ \hline 174 \\ 9 \\ \hline 165. \end{array}$$

$$\Rightarrow \boxed{\sqrt{\frac{35 \pm \sqrt{35^2 - 4\cdot 165}}{2}} = \sigma_1, \sigma_2.}$$

4

165 | 5
15 | 33
15

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$35^2 - 4 \cdot 165 = 5^2 \cdot 7^2 - 2^2 \cdot 5 \cdot 3 \cdot 11$

$= 5(5 \cdot 7^2 - 4 \cdot 3 \cdot 11)$.

$= 5 \cdot 113 = 565$.

$\Downarrow$

$$\boxed{\sigma_{1,2} = \frac{35 \pm \sqrt{565}}{2}}$$

4
49
5
245
11
12
132.

245
132
113

7. **[10 points]** Compute the positive singular values of the following matrix:

$$A = \left(\begin{array}{ccccc|ccc} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right) = \begin{pmatrix} \tilde{A} & 0 \\ 0 & 0 \end{pmatrix}$$

$(5,5)$ $(5,3)$

$$AA^T = \begin{pmatrix} \tilde{A}^2 & 0 \\ 0 & 0 \end{pmatrix} = \left(\begin{array}{ccccc|c} 1 & & & & & \\ & 2^2 & & & 0 & \\ & & 1 & & & \\ & & & 3^2 & & 0 \\ & 0 & & & 2^2 & \\ \hline & & & 0 & & 0 \end{array}\right)$$

$(5,5)$ $(5,5)$

$$\Rightarrow \boxed{\sigma_1 = 3 > \sigma_2 = \sigma_3 = 2 > \sigma_4 = \sigma_5 = 1}$$

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